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The Role of Route-Level Decisions in the Efficiency and Resilience of Airline Operations: Evidence from the Wright Amendment Repeal

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Abstract. *Problem definition:* We study the impact of an airline’s route-level decisions on the efficiency and resilience of its operations. *Methodology/results:* Our study uses a proprietary passenger-level data set provided by Southwest Airlines and takes advantage of a regulatory change, the Wright Amendment repeal. The Wright Amendment restricted nonstop service between Dallas Love Field Airport (DAL) and most major destinations. Its repeal led to significant changes in Southwest’s service offerings at DAL. We use a difference-in-differences framework to quantify the impact of the repeal on the scheduled travel times and long travel delays at the passenger level, which serve as proxies for efficiency and resilience, respectively. We show that the repeal’s impact varies widely across destinations. Although Southwest introduced nonstop service to some destinations but not others after the repeal, this decision alone does not sufficiently explain the observed heterogeneity. As such, we develop two route-level metrics—route inefficiency and route resilience—to explain the heterogeneous impact of the repeal on different destinations. Moreover, our counterfactual analysis shows that reallocating connecting passengers with small itinerary buffers to alternative itineraries can result in major resilience improvements without significantly deteriorating efficiency. *Managerial implications:* Our findings are robust to alternative empirical specifications and have important implications for practitioners and regulators. Specifically, our study not only sheds light on the operational consequences of airlines’ route-level decisions but also demonstrates how managers can effectively manage the efficiency-resilience tradeoff for connecting passengers. Our study also implies that regulators can add more transparency to airline operations by providing public access to passenger-level data.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/msom.2023.0070>.

Keywords: airline operations • efficiency • resilience • travel times • travel delays

1. Introduction

According to the Federal Aviation Administration (FAA 2022), the aviation industry contributes more than \$500 billion to the gross domestic product (GDP) of the United States. Despite this contribution, operational disruptions in the airline industry (e.g., flight delays) cost airlines and passengers billions of dollars every year (Arkan et al. 2013, Ferguson et al. 2013). Indeed, the FAA (2020) estimates that the hourly value of travel time savings for passengers is \$36.10 for personal travel and \$63.20 for business travel. Given the critical economic role of the airline industry and the benefits of travel time savings for passengers, it is important to understand the impact of airlines’ route planning and flight scheduling decisions on passenger travel times and delays. These decisions include whether to offer

nonstop service between two airports, which layover airports to use for connecting passengers, and how to set flight times and frequencies. Hereinafter, we refer to an airline’s route planning and flight scheduling decisions as route-level decisions for brevity.

The airline industry studies focusing on travel times and delays typically use the Airline On-Time Performance (OTP) data published by the Bureau of Transportation Statistics (BTS; e.g., Ramdas et al. 2013, Wang et al. 2022). OTP data contain scheduled and actual departure and arrival times, allowing scholars to analyze travel times and delays at the flight level. However, OTP data lack passenger-level details (e.g., passenger itineraries showing flight information and layover times). This limitation makes capturing passengers’ travel experiences with OTP data challenging, especially when they

have connecting flights (Barnhart et al. 2014, Guo et al. 2022). We overcome this limitation by using a proprietary passenger-level data set provided by Southwest Airlines.¹

1.1. Novelty of Passenger-Level Data

Our primary data source is a data set from Southwest Airlines that spans the three-year period between January 1, 2013, and December 31, 2015, and includes itinerary details for all passengers originating from Dallas Love Field Airport (DAL) and Houston Hobby Airport (HOU).² The data set contains itinerary records of approximately 20 million passengers. Table 1 illustrates the structure of our data set with itinerary records of three passengers. The first passenger (i.e., Passenger ID = 1) took a nonstop flight from DAL to Austin (AUS) on May 30, 2014. The second and third passengers shared the same flight, both having a layover in AUS, with final destinations of Las Vegas (LAS) and Denver (DEN), respectively.

Although OTP data contain the scheduled and actual gate departure and gate arrival times of all three flights we list in Table 1, they do not include passenger itinerary information. Thus, OTP data are uninformative about the number of passengers traveling from DAL to AUS, LAS, and DEN and their scheduled travel times (STTs), which include layover times at AUS for the second and third passengers. In addition, as Table 1 shows, the flight from DAL to AUS had a 36-minute arrival delay as the scheduled and actual gate arrival times were 18:45 and 19:21, respectively. This delay was unlikely to harm the second passenger as they had ample time to catch their connecting flight and reach LAS on time. By contrast, the third passenger missed their connection in AUS because the first leg’s actual gate arrival time (19:21) was later than the second leg’s actual gate departure time (19:18). OTP data cannot capture the stark contrast between the second and third passengers’ travel experiences because of the absence of passenger-level information. Section A.1 in the online appendix provides a more detailed comparison between our and publicly available data sets.

1.2. Wright Amendment and Identification Strategy

Airlines typically make route-level decisions to respond to (or in anticipation of future) passenger demand.

Specifically, a point-to-point airline, such as Southwest, is more likely to introduce nonstop service from an origin airport o to a destination airport d when the $o - d$ market demand (i.e., the number of passengers who want to travel from o to d) is high (e.g., Boguslaski et al. 2004). By contrast, a hub-and-spoke airline, such as United Airlines, typically connects o to a hub airport h if there is enough demand from o to h and other airports that can be reached with a layover at h (see Belobaba et al. 2015 for an overview of airlines’ route-level decisions). Under typical operational conditions, an airline’s route-level decisions evolve gradually. Consequently, it can be challenging to disentangle the impact of route-level decisions from other managerial choices that also change over time (e.g., changes in loyalty programs or seating policy). By contrast, our data set covers a period during which a regulatory change, the repeal of the Wright Amendment, allowed Southwest to swiftly make major route-level changes for its DAL-origin passengers.

The 1979 Amendment to the Federal Aviation Act of 1958, widely known as the Wright Amendment, restricted nonstop service from DAL to stimulate the growth of the Dallas/Fort Worth Airport (DFW). Initially, the Wright Amendment allowed nonstop service between DAL and airports only within Texas and its four neighboring states (Arkansas, Louisiana, New Mexico, and Oklahoma). Over time, several modifications allowed nonstop service between DAL and destinations in Alabama, Kansas, Mississippi, and Missouri. The Wright Amendment was repealed on October 13, 2014. Figure 1, (a) and (b), shows the destinations to which Southwest was offering nonstop service from DAL in the one-year period before the repeal and the one-year period after the repeal, respectively. For comparison purposes, Figure 1, (c) and (d), shows Southwest’s nonstop service offerings from Houston Hobby Airport in the same time frame. These figures show that the nonstop service offerings from DAL and HOU were dramatically different before the repeal but became almost identical afterward.

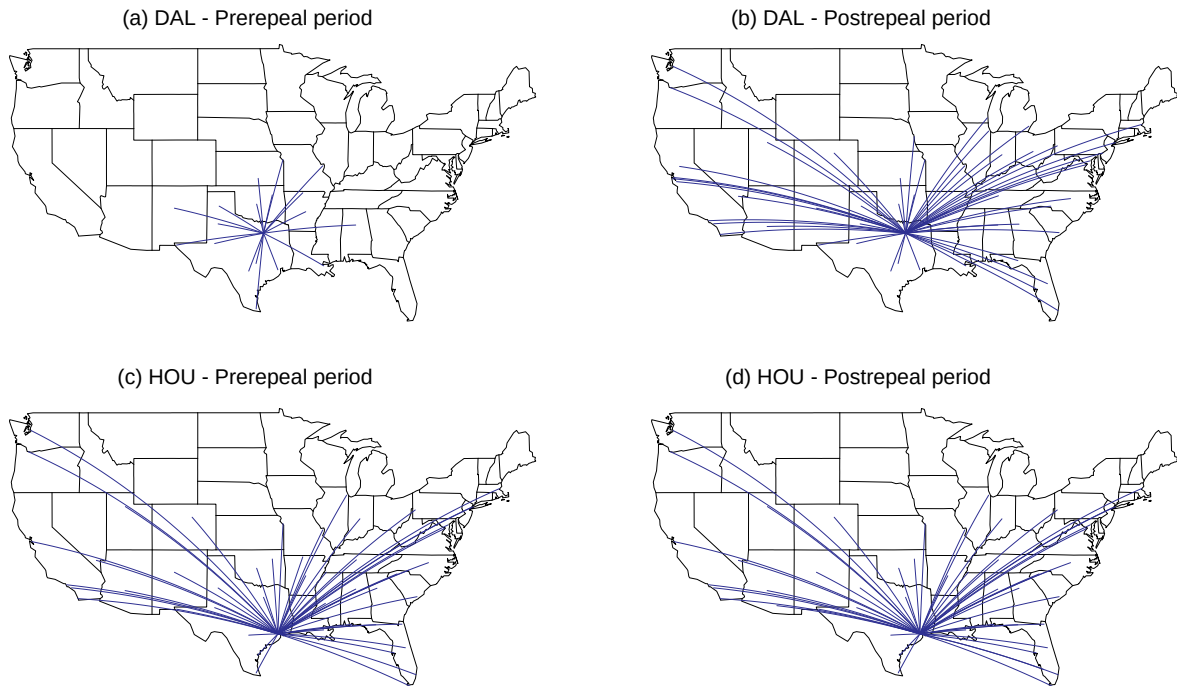
We leverage the route-level changes triggered by the Wright Amendment repeal to answer two research questions with important practical implications. First, how do an airline’s route-level decisions affect the efficiency and resilience of its operations for its passengers?

Table 1. Three Example Itineraries

Departure date	Passenger ID	Leg	Origin	Destination	SDT	ADT	SAT	AAT
May 30, 2014	1	1	DAL	AUS	17:50	18:24	18:45	19:21
May 30, 2014	2	1	DAL	AUS	17:50	18:24	18:45	19:21
May 30, 2014	2	2	AUS	LAS	20:25	20:21	21:20	21:19
May 30, 2014	3	1	DAL	AUS	17:50	18:24	18:45	19:21
May 30, 2014	3	2	AUS	DEN	19:20	19:18	20:35	20:35

Note. In each row, SDT, ADT, SAT, and AAT are the scheduled gate departure time, actual gate departure time, scheduled gate arrival time, and actual gate arrival time of the corresponding flight, respectively.

Figure 1. (Color online) Southwest's Nonstop Service Offerings from DAL and HOU



To understand the impact of an airline's decisions on its passengers' travel experiences, we use scheduled travel times and long travel delays (delays exceeding three hours) at the passenger level as proxies for efficiency and resilience, respectively. Second, how can an airline improve the resilience of its operations for its connecting passengers without significantly compromising its efficiency?

In light of Figure 1, we adopt a difference-in-differences approach with HOU as a control group. We use HOU as a control group for four other reasons: (i) both Dallas and Houston are in Texas with similar demographics and economic conditions; (ii) both cities have major international airports, DFW and George Bush Intercontinental Airport (IAH), respectively; (iii) during our study period, Southwest served only the secondary airports in both cities, where it was the dominant airline in both DAL and HOU; and (iv) senior leadership at Southwest confirmed our choice of HOU as a control group. From an econometric standpoint, using HOU as a control group requires the parallel trends assumption and the stable unit treatment value assumption (SUTVA). To this end, we conduct extensive empirical tests in Section 6 to validate our choice of HOU as a control group and demonstrate that our results remain robust when we account for potential violations of the difference-in-differences model assumptions.

1.3. Operational Efficiency

Several research streams, including supply chain management, logistics, and transportation studies, focus on

quantifying and improving the efficiency of transportation systems (e.g., Simchi-Levi et al. 2005, Liu and Hansen 2019). Specifically, the transportation literature relies on a transportation system's spatial properties to develop efficiency metrics, such as (i) the inverse of the distance between the two nodes (e.g., Latora and Marchiori 2001); (ii) the mean ratio of the Euclidean distance between the two nodes and the distance along edges passengers use to travel from one node to the other (e.g., Louf et al. 2013); and (iii) the difference between the mean commute distance and the theoretical minimum commute distance (e.g., Kanaroglou et al. 2015). Although these metrics serve as proxies for a ground transportation system's efficiency, they do not capture essential features of air transportation. For example, airlines determine flight times based on airport congestion (Nicolae et al. 2017). Furthermore, passengers with connecting flights incur layover times. Lastly, although airlines have been increasing scheduled flight times, consumers still prefer shorter scheduled travel times (e.g., Forbes et al. 2019, Yimga and Gorjidoz 2019). Accordingly, we use passengers' scheduled travel times as a proxy for the efficiency of an airline's operations. For each passenger, we measure the scheduled travel time as the difference between the scheduled arrival time of the last leg and the scheduled departure time of the first leg in the passenger's itinerary.

Twenty destinations in Southwest's network were exempt from the Wright Amendment. Among the remaining 68 destinations subject to the Wright Amendment, the impact of the repeal on passengers traveling

to a specific destination may depend on whether Southwest introduced nonstop service to that destination after the repeal. Accordingly, we classify the destinations in Southwest’s network into three categories: (i) type I destinations are those exempt from the Wright Amendment, (ii) type II destinations are nonexempt destinations to which Southwest did not introduce nonstop service after the repeal, and (iii) type III destinations are nonexempt destinations to which Southwest introduced nonstop service after the repeal. Section A.2 in the online appendix lists the type I, II, and III destinations in Southwest’s network during our study period.

As an example, Figure 2 shows the daily average STTs from DAL and HOU to Chicago-Midway (MDW), a type III destination, during the two-year period between October 13, 2013, and October 12, 2015. Before the repeal, Southwest transported passengers from DAL to MDW through a layover in an exempt airport, such as Kansas City (MCI) or St. Louis (STL). Southwest introduced nonstop service from DAL to MDW on the day of the repeal, reducing the average STT from 209 minutes to 139 minutes. By contrast, the average STTs from HOU to MDW were 148 and 152 minutes in the pre- and postrepeal periods, respectively.

Table 2 provides further examples by showing the average STTs from DAL and HOU to six destinations. The changes in the average STTs for passengers traveling from DAL to Albuquerque (ABQ) and MCI were relatively small (2.44 minutes and 0.07 minutes, respectively) because almost all passengers traveling from DAL to these two type I destinations took nonstop flights during both pre- and postrepeal periods. Although Southwest did not introduce nonstop service to type II destinations Albany (ALB) and Minneapolis (MSP), the average STT from DAL to ALB decreased by 61 minutes, whereas the average STT from DAL to MSP increased by 35 minutes. Southwest introduced nonstop service from DAL to type III destinations MDW and Santa Ana (SNA), significantly reducing the average STTs (143 minutes for SNA and 70 minutes for MDW).

Although these examples validate the expected efficiency advantage of nonstop flights, they also illustrate that destination type is insufficient to explain the sign and magnitude of the changes in efficiency.

We construct a series of models to unveil the impact of route-level decisions on operational efficiency. After controlling for several factors (e.g., weather conditions, daily and monthly seasonality), we show that the average reductions in STT to type I, type II, and type III destinations were 5, 32, and 79 minutes, respectively. We then introduce our route inefficiency metric, which combines travel distances and airport congestion, to capture the heterogeneity in STT changes among destinations. Lastly, we perform destination-level efficiency analyses to revisit the examples in Table 2 and explain why Southwest’s route-level decisions benefited some destinations more than others (e.g., SNA versus MDW) and even hurt some destinations (e.g., MSP).

1.4. Operational Resilience

The broader operations literature defines resilience as the ability to recover from disruptions and disasters (e.g., Sheffi 2007). According to an Oliver Wyman report (Stalnaker et al. 2019, p. 32), in the airline industry, “operational resilience is the ability of airlines to adhere to the schedules they publish, allowing for potential disruption and building in a cushion to recover when the unexpected happens.” The same report (page 32) contends that “the more capacity that airlines add, the more that bottlenecks in the system worsen and the more that airlines’ operational resilience is tested.” The Wright Amendment repeal provides a practical setting to study the role of an airline’s route-level decisions in the resilience of its operations. On the one hand, the repeal might harm the resilience of Southwest’s operations because Southwest started offering more flights out of DAL after the repeal, leading to more air traffic and ground congestion at DAL. On the other hand, the repeal enabled Southwest to offer nonstop service from DAL to many destinations, where switching

Figure 2. (Color online) Daily Average Scheduled Travel Times from DAL and HOU to MDW

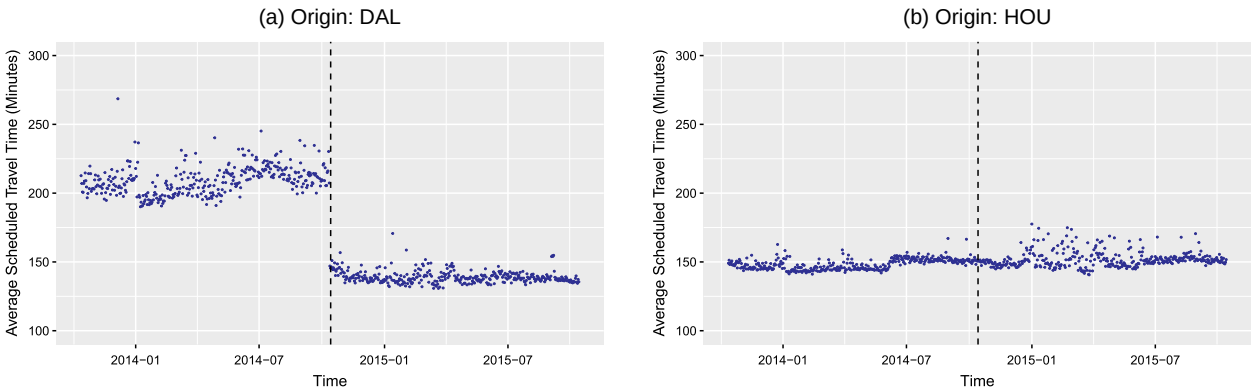


Table 2. Average Scheduled Travel Times to Six Destinations

Destination	Destination type	DAL-origin passengers			HOU-origin passengers		
		Pre	Post	Difference	Pre	Post	Difference
ABQ	Type I	107.76	110.20	2.44	142.79	149.54	6.75
MCI	Type I	86.65	86.72	0.07	148.37	143.59	−4.78
ALB	Type II	405.24	344.15	−61.09	356.30	357.39	1.09
MSP	Type II	236.26	271.37	35.11	304.38	301.12	−3.26
MDW	Type III	209.26	139.15	−70.11	148.33	151.89	3.56
SNA	Type III	362.26	219.10	−143.16	237.16	250.52	13.36

from connecting flights to nonstop flights can improve resilience by eliminating one flight leg and potential disruptions that can occur during the layover. Thus, the resilience impact of these changes is not fully clear, and it is important to understand the overall effect of these competing forces.

Both academics and practitioners have commonly used on-time performance (i.e., the percentage of flights arriving less than 15 minutes after the scheduled arrival time) as a proxy for an airline's ability to adhere to its planned schedule (e.g., Prince and Simon 2009, Stalnakker et al. 2019). Despite its frequent use, on-time performance has shortcomings partly because airlines have been increasing their scheduled flight times (referred to as schedule padding) to improve on-time performance (Forbes et al. 2019). Indeed, Ramdas et al. (2013) show that long delays (measured in several different ways, including delays exceeding two hours, three hours, and three standard deviations above the mean delay) reduce contemporaneous stock returns, whereas late flights (more than 15 minutes late) do not affect stock returns. Similarly, Alan and Lapré (2018) show that long delays are associated with future financial distress, whereas on-time performance has no such association. A potential explanation for these findings is that long delays can lead to missed connections and create additional operational expenses (e.g., extra crew and fuel costs and passenger compensation) for airlines (e.g., Ferguson et al. 2013, Marla et al. 2017). However, on-time performance cannot capture the impact of a long delay on connecting passengers' subsequent flights because it is tracked at the flight rather than the passenger level.

Considering the shortcomings of on-time performance at the flight level, we focus on long travel delays at the passenger level and use a binary classification to label a passenger as delayed if the passenger's travel delay exceeds three hours. We show that the repeal increased the likelihood of a passenger experiencing a long delay while traveling from DAL to the exempt (type I) destinations while improving resilience for passengers traveling from DAL to the nonexempt (type II and type III) destinations. Additionally, we develop a route resilience metric, which combines itinerary buffers and flight frequencies, to unveil the mechanism behind

the heterogeneous impact of the repeal across destinations. Lastly, our destination-level resilience analysis explains why the route-level changes triggered by the repeal benefited some destinations and hurt others.

1.5. Managing the Efficiency-Resilience Tradeoff

Several studies show the importance of air and ground buffers for reducing flight delays (e.g., Ahmadbeygi et al. 2010, Deshpande and Arkan 2012, Brueckner et al. 2021). Although these studies are informative for nonstop passengers, they do not fully capture connecting passengers' travel experiences. By contrast, our passenger-level data set enables us to conduct a counterfactual analysis with a numerical optimization procedure to shed light on the relationship between efficiency and resilience for connecting passengers. Our procedure identifies the passengers with the shortest itinerary buffers and reallocates them to alternative itineraries with longer buffers. We show that our procedure can lead to a major improvement in resilience without significantly harming efficiency. For example, reallocating 2% of the connecting passengers with the shortest buffers to alternative itineraries would reduce the number of connecting passengers experiencing long delays by 12% while increasing the average scheduled travel time by only 0.33% (0.9 minutes).

In summary, we contribute to the literature by leveraging the Wright Amendment repeal to quantify the impact of an airline's route-level decisions on the efficiency and resilience of its operations and by showing how an airline can manage the efficiency-resilience tradeoff for its connecting passengers. The rest of the paper is organized as follows. Section 2 describes our data set and variables. Sections 3 and 4 present our efficiency and resilience analysis, respectively. Section 5 presents our counterfactual analysis, Section 6 presents our robustness checks, and Section 7 concludes by discussing the implications of our findings for regulators, managers, and researchers.

2. Data and Variables

In our main analysis, we use a subset of the data set from Southwest Airlines covering the period between

October 13, 2013, and October 12, 2015 (i.e., one year before and one year after the repeal). We use the remaining data for robustness checks. We combine our main data set with three publicly available data resources to construct our explanatory and control variables. Specifically, we use OTP data and the FAA's Operations and Performance Data, which comprise the departure and arrival activities of the 77 major U.S. airports, to develop our route-level metrics. In addition, we use the Iowa State University Automated Airport Weather Observations Database to construct our weather control variables. Section A.1 in the online appendix provides more details about the publicly available data sets we use in our study.

2.1. Dependent Variables

We use scheduled travel times to quantify the efficiency impact of the Wright Amendment repeal. Let L_i denote the number of legs in passenger i 's itinerary, and let SDT_{il} and SAT_{il} denote the scheduled gate departure and gate arrival times of leg $l = 1, \dots, L_i$, respectively. Then, after adjusting for the time zone difference, the scheduled travel time for passenger i is $STT_i = SAT_{i,L_i} - SDT_{i,1}$. This formula allows us to incorporate layover times into our efficiency analysis. For example, revisiting Table 1, the scheduled travel time for the second passenger is 5.5 hours (with an hour-and-40-minute layover in AUS).

We use long travel delays to quantify the resilience impact of the repeal. Specifically, we construct a binary variable $Delayed_i$ that equals one if passenger i 's arrival delay to their final destination exceeds three hours. To this end, let AAT_{i,L_i} denote the actual arrival time of the last leg in passenger i 's itinerary. When a passenger completes their scheduled flight(s), $Delayed_i = 1$ if $AAT_{i,L_i} - SAT_{i,L_i} > 3$ hours. In some cases, a passenger's itinerary is *disrupted* because of a missed connection. Our data set does not specify the disrupted passengers due to missed connections or how they were rebooked on subsequent flights. Therefore, we first identify the disrupted passengers, reallocate them to available subsequent flights from the point of disruption to their final destinations, and then calculate their travel delays based on their new itineraries. Formally, let $AAT_{i,\hat{L}_i}$ denote the actual arrival time of the flight passenger i takes to their final destination after rebooking. Then, for a disrupted passenger, $Delayed_i = 1$ if $AAT_{i,\hat{L}_i} - SAT_{i,L_i} > 3$ hours.

We identify disrupted passengers by following the approach of Herring et al. (2019) to specify airport-specific minimum layover times. We extract each airport's minimum scheduled layover time from our passenger-level data set. Let $MSLT_a$ denote the minimum scheduled layover time for airport a . Even though there are a few scheduled layover times that are shorter than 20 minutes, to be conservative in our estimates, we assume that a passenger misses a connection if the

available layover time (i.e., the difference between the actual arrival time of the disrupted leg and the actual departure time of the next leg) is less than 20 minutes. As a result, we assume an itinerary with multiple legs is disrupted because of a missed connection when the available layover time is less than $\max\{20, MSLT_a\}$. The next step in our displacement process is to reallocate the disrupted passengers. To accomplish this, we search for flights departing from the airport where the disruption occurs and allocate a disrupted passenger to the next available flight to their final destination. We provide a more detailed description of our displacement procedure in Section A.3 in the online appendix.

The proportion of disrupted passengers and the travel delay of a disrupted passenger depend on the displacement procedure we follow. Thus, to ensure that our resilience results are robust to our displacement procedure, we rerun our analysis using (i) a two-hour cutoff to define our long delay variable (i.e., redefine $Delayed_i$ so that it equals one if passenger i 's delay exceeds two hours and zero otherwise) and (ii) the actual delay in minutes as our dependent variable. See Section 6 for our robustness checks.

2.2. Explanatory and Control Variables

Regarding our explanatory variables, $Dallas_i$ is a binary variable indicating whether passenger i 's origin was DAL, and $Post_i$ is a binary variable indicating whether passenger i 's itinerary date was after the repeal. Furthermore, we use the interaction of these two variables, $Treated_i$ ($Dallas_i \times Post_i$), to measure the impact of the repeal on DAL-origin passengers.

Adverse weather conditions can negatively affect travel times. Airline studies using flight-level data typically include separate weather controls for each flight's origin and destination airports. For example, Nicolae et al. (2017) use the minimum temperatures at the origin and destination airports on the day of a particular flight as two separate control variables. Because we use passenger-level data, some passengers in our data set have nonstop flights, whereas others have one or more layovers. As such, we use the minimum temperature during a passenger's trip as a single temperature control variable. For example, suppose that a passenger travels from DAL to MDW through a layover in MCI and that the minimum temperatures at DAL, MCI, and MDW on the day of that passenger's trip are 55, 45, and 30° Fahrenheit, respectively. Then, the temperature control variable for that passenger equals $\min\{55, 45, 30\} = 30$. Similarly, we include the maximum wind, maximum precipitation, and minimum visibility as additional controls. Our control variables also include the nonstop distance between the origin and destination airports and destination, month, and day of the week fixed effects.

Increased airport congestion can result in longer travel times. Therefore, measuring the airport congestion at

each leg of a passenger's trip is important. We measure the congestion at the departure and arrival airports for each leg in a passenger's itinerary as follows: Let DC_{il} denote the departure congestion of leg l in passenger i 's itinerary. To calculate DC_{il} , we first count the number of flights scheduled to depart between 45 minutes before and 15 minutes after the scheduled departure time of that flight using OTP data, which allow us to consider all flights from all airlines scheduled to depart from the origin airport of that flight. We then obtain the average hourly departure capacity of the origin airport from the FAA's Operations and Performance Data. Lastly, we calculate DC_{il} as the ratio of the number of departures we count within the one-hour time block to the average hourly departure capacity of the departure airport. This three-step procedure is identical to the one used by Nicolae et al. (2017). We calculate the arrival congestion of leg l for passenger i , AC_{il} , similarly by using the total number of arrivals to the destination airport of leg l within the one-hour block around leg l 's scheduled arrival time and the average hourly arrival capacity of the destination airport. We use the maximum congestion at the time of each passenger's trip, that is, $\max\{AC_{i1}, \dots, AC_{i,L_i}, DC_{i1}, \dots, DC_{i,L_i}\}$, as a single control variable.

We use two additional variables to construct the route resilience metric. First, we calculate the nominal travel time for each passenger. Airlines typically adjust their flight times by inserting buffers into their schedules to mitigate the impact of delays. These flight buffers can be in the form of flight and ground buffers (Brueckner et al. 2021). Flight buffers equal the time added to the nominal gate-to-gate travel time to compensate for expected delays during taxiing and flying. Similarly, ground buffers equal the time added to the nominal ground time to compensate for expected delays during layovers. Following Kafle and Zou (2016), we consider the 10th percentile values of gate-to-gate scheduled flight times and scheduled layover times to calculate the nominal travel time for passenger i , $NomTT_i$. Second, our route resilience metric also includes flight frequencies. More formally, we define $flight_{il}$ as the number of nonstop flights from the origin to the destination of leg l on passenger i 's travel date. Section A.4 in the online appendix provides more detailed descriptions of the key variables in our study along with some numerical examples.

2.3. Summary Statistics

In the prerepeal period, 48.4% of DAL-origin passengers had a nonstop flight, 45.4% had one layover, and 6.2% had two or more layovers. In the postrepeal period, 78.9% of DAL-origin passengers had a nonstop flight, 20.4% had one layover, and 0.7% had two or more layovers. For comparison purposes, the proportion of HOU-origin passengers who had nonstop flights was 78.4% before the repeal and 75.1% after. These statistics

support our assertion that the Wright Amendment forced Southwest to deviate from its point-to-point model in DAL. Additionally, the proportion of DAL-origin passengers experiencing long delays was 2.5% before the repeal and 1.23% after. Although these proportions seem small, their difference implies a sizable reduction in the number of passengers with long delays. For comparison, the proportion of HOU-origin passengers experiencing long delays was 1.67% before the repeal and 1.33% after. Table 3 provides additional summary statistics. Notably, DAL-origin passengers had a shorter average STT after the repeal (173 versus 160 minutes), whereas the average nonstop distance was longer (696 versus 816 miles) because of the introduction of long-haul flights. The distributions of other variables listed in Table 3 also show major changes for DAL-origin passengers after the repeal. By contrast, the variable distributions for HOU-origin passengers show smaller changes over time (e.g., the average nonstop distance was 921 miles before the repeal and 951 miles after), providing additional support for the validity of HOU as a control group. Table A4 in the online appendix provides summary statistics for the variables we construct in Sections 3 and 4.

3. Efficiency Analysis

3.1. Analysis by Destination Type

In this section, we first use the following model specification to quantify the overall efficiency impact of the repeal on DAL-origin passengers:

$$STT_i = \alpha_{0,d(i)} + \alpha_1 Dallas_i + \alpha_2 Post_i + \alpha_3 Treated_i + \gamma X_i + \epsilon_i. \quad (1)$$

In Equation (1), STT_i is the scheduled travel time for passenger i , $\alpha_{0,d(i)}$ is destination fixed effect for passenger i 's final destination (i.e., $d(i)$), X_i is a column vector that includes control variables, γ is a row vector that represents the coefficients of those control variables, and ϵ_i is the error term. In our analyses, we cluster standard errors at the destination level.

Table 4 shows our coefficient estimates. Model 1 contains approximately 12.5 million passengers who traveled from DAL and HOU to all other airports in Southwest's network.³ The treatment effect coefficient in Model 1, α_3 , suggests that the repeal resulted in a 41-minute decrease in STT for DAL-origin passengers, on average. Although the magnitude of this decrease indicates a sizable overall efficiency gain, it does not reveal which destinations benefited the most from the repeal. To this end, our next model estimates the differential impact of the repeal on the three destination types. Specifically, let $TypeII_i$ denote an indicator variable that equals one if passenger i 's destination is a type II airport and zero otherwise. Similarly, $TypeIII_i$ is an indicator variable that equals one if passenger i 's

Table 3. Summary Statistics

Metric	DAL prerepeal							DAL postrepeal						
	Mean	SD	P10	P25	Median	P75	P90	Mean	SD	P10	P25	Median	P75	P90
Scheduled travel time	172.66	108.67	60	75	170	250	320	159.53	85.33	65	90	145	200	285
Nominal travel time	144.52	86.31	51	62	135	207	262	135.59	66.16	54	81	129.90	175.80	227
Travel delay	28.12	125.13	−11	−5	4	21	54	12.19	78.96	−16	−9	−2	12	39
Flights	7.33	3.53	3	5	6.50	9	13	5.03	2.71	2	3	4	6	10
Nonstop distance	695.69	391.90	248	323	585	973	1,246	815.69	391.71	293	460	793	1,182	1,360
Min. temperature	49.72	18.60	24.98	35.06	51.98	66.02	71.96	52.39	17.53	28.04	39.02	55.04	66.20	73.94
Max. wind	18.07	5.10	12	14	18	21	24	16.71	5	11	13	16	19	23
Max. precipitation	0.09	0.23	0	0	0	0.06	0.27	0.11	0.26	0	0	0	0.09	0.33
Min. visibility	5.22	3.49	0.50	1.75	5	9	10	5.40	3.77	0.50	1.50	6	10	10
Max. congestion	0.39	0.16	0.20	0.27	0.36	0.47	0.61	0.46	0.16	0.27	0.35	0.43	0.53	0.67

Metric	HOU prerepeal							HOU postrepeal						
	Mean	SD	P10	P25	Median	P75	P90	Mean	SD	P10	P25	Median	P75	P90
Scheduled travel time	170.49	88.34	65	110	150	210	305	179.72	90.77	75	115	155	220	320
Nominal travel time	147.23	70.80	55	99	132	189.30	254	153.09	70.06	70	103	137	193	256.80
Travel delay	19.66	88.69	−12	−6	3	20	51	12.46	78.95	−16	−10	−2	12	41
Flights	4.52	2.96	2	3	4	5	8	4.60	2.67	2	3	4	5.50	8.50
Nonstop distance	921.19	427.82	302	571	883	1,246	1,428	951.33	413.59	393	670	937	1,246	1,428
Min. temperature	51.34	17.86	26.96	39.02	53.96	66.02	73.04	52.96	17.22	28.94	41	55.94	66.92	73.04
Max. wind	16.53	5.01	11	13	16	19	23	16.02	4.82	11	13	15	19	22
Max. precipitation	0.10	0.22	0	0	0.01	0.09	0.30	0.13	0.27	0	0	0.01	0.13	0.38
Min. visibility	5.34	3.78	0.50	1.50	6	10	10	5.24	3.85	0.50	1.25	5	10	10
Max. congestion	0.48	0.15	0.32	0.38	0.46	0.57	0.68	0.48	0.16	0.29	0.37	0.46	0.57	0.69

Notes. We report the mean, standard deviation (SD), and 10th (P10), 25th (P25), 50th (median), 75th (P75), and 90th (P90) percentiles of each variable for each origin-period combination. Scheduled travel time, nominal travel time, and travel delay are in minutes. Flights show the summary statistics for $flight_i \equiv (\sum_{l=1}^{L_i} flights_{il})/L_i$. Nonstop distance is in miles, min. temperature is in Fahrenheit, max. wind is in knots, max. precipitation is in inches/hour, min. visibility is in miles, and max. congestion is a scalar between zero and one.

destination is a type III airport and zero otherwise. We add the interactions of $TypeII_i$ and $TypeIII_i$ with $Dallas_i$, $Post_i$, and $Treated_i$ to Equation (1) to specify Model 2.

In Model 2, $\hat{\alpha}_3 = -4.66$ captures the impact of the repeal on type I destinations. Because almost all passengers traveling from DAL and HOU to type I destinations take nonstop flights, the 4.66-minute decrease in STT implies that Southwest slightly reduced the scheduled flight times from DAL to those destinations. This observation is consistent with the finding that airlines allocate longer buffers to flights with more connecting passengers (e.g., Baumgarten et al. 2014, Kafle and Zou 2016). In other words, the repeal decreased the number of DAL-origin passengers using the exempt destinations as layover airports, which, in turn, allowed Southwest to reduce the buffer times for flights from DAL to those destinations. The α_8 coefficient in Model 2 indicates that the decrease in STT is 26.97 minutes more for type II destinations. In other words, the repeal led to a 31.63-minute decrease in STT to type II destinations (i.e., $\hat{\alpha}_3 + \hat{\alpha}_8 = -31.63$). Similarly, the repeal resulted in a 78.62-minute reduction in STT to type III destinations (i.e., $\hat{\alpha}_3 + \hat{\alpha}_9 = -78.62$). Furthermore, comparing α_8 and α_9 with an F -test reveals a statistically significant difference between the reductions in STT to type II and type III destinations ($p < 10^{-10}$). These findings imply that

type I (type III) destinations had the smallest (largest) efficiency gains from the repeal, on average.

Although Model 2 quantifies the average impact of the repeal on different airport groups, it does not reveal why some destinations benefited more than others within the same group (e.g., ALB versus MSP among type II destinations). In the next section, we propose a *route inefficiency* metric to shed light on the heterogeneous impact of the repeal on different destinations.

3.2. The Role of Route Inefficiency

The transportation literature (e.g., Louf et al. 2013, Dong et al. 2016, Liu and Hansen 2019) indicates that the efficiency of a route depends on (i) the total distance passengers travel along the route relative to the shortest distance between their origin and destination and (ii) the congestion they face along the route. Using publicly available data, we operationalize these two factors as follows: Let $distance_{il}$ denote the distance between the origin and destination airports in leg l , and let $non.stop.distance_i$ denote the nonstop distance between passenger i 's origin and destination. Furthermore, recall that DC_{il} and AC_{il} denote the departure and arrival congestion for leg l in passenger i 's itinerary, respectively. Then, we define the route

Table 4. Efficiency Regressions

	Model 1	Model 2	Model 3
Intercept	−53.87939*** (0.19245)	−38.66395*** (0.18013)	−41.30829*** (0.18376)
Nonstop distance	0.21390*** (0.00022)	0.23286*** (0.00021)	0.21470*** (0.00021)
Min. temperature	−0.30088*** (0.00165)	−0.22539*** (0.00157)	−0.27486*** (0.00159)
Max. wind	0.51571*** (0.00293)	0.34348*** (0.00272)	0.43412*** (0.00280)
Max. precipitation	1.11562*** (0.06631)	1.44115*** (0.06287)	1.02546*** (0.06431)
Min. visibility	−0.75529*** (0.00406)	−0.62625*** (0.00380)	−0.69133*** (0.00391)
Max. congestion	56.92729*** (0.10265)	41.24650*** (0.10149)	46.38823*** (0.10149)
Dallas, α_1	49.67712*** (0.04375)	0.65176*** (0.04484)	27.28900*** (0.04515)
Post, α_2	6.44850*** (0.03555)	2.94791*** (0.05280)	4.62313*** (0.04355)
Treated (Dallas × Post), α_3	−40.99425*** (0.05209)	−4.65915*** (0.05979)	−17.67652*** (0.05689)
Dallas × Type II, α_4		74.97254*** (0.19333)	
Dallas × Type III, α_5		89.60072*** (0.07035)	
Post × Type II, α_6		7.85286*** (0.14796)	
Post × Type III, α_7		5.32730*** (0.06858)	
Treated × Type II, α_8		−26.97086*** (0.26862)	
Treated × Type III, α_9		−73.96227*** (0.09316)	
Dallas × ACRI, α_{10}			−123.17100*** (0.13348)
Post × ACRI, α_{11}			−8.21067*** (0.11776)
Treated × ACRI, α_{12}			128.73220*** (0.17362)
Observations	12,468,784	12,468,784	12,416,724
Adjusted R^2	0.77090	0.80005	0.78991

Notes. Standard errors clustered at the destination level are in parentheses. The dependent variable is the scheduled travel time in all three models. All models have destination, month, and day-of-the-week fixed effects. We do not report the fixed effect coefficients because of space limitations.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

inefficiency for passenger i as

$$RI_i = \frac{\sum_{l=1}^{L_i} \text{distance}_{il}}{\text{non.stop.distance}_i} \prod_{l=1}^{L_i} \left(1 + \frac{DC_{il} + AC_{il}}{2} \right). \quad (2)$$

In Equation (2), having multiple legs in the itinerary and traveling through more congested airports increase a passenger's route inefficiency score.⁴ We expect a decrease in the route inefficiency scores of DAL-origin passengers after the repeal because of the availability of nonstop service to many destinations. For example, if Southwest had a nonstop flight from HOU to a type III destination before the repeal, introducing nonstop

service from DAL to the same destination would lead to a large decrease in RI for DAL-origin passengers, whereas RI for HOU-origin passengers would remain stable. Indeed, we predict that the destination-level changes in RI values would explain the heterogeneity of the efficiency changes across destinations. To this end, let $\overline{RI}_{o,d,pre}$ and $\overline{RI}_{o,d,post}$ denote the average RI values for the set of passengers who traveled from origin $o \in \{DAL, HOU\}$ to destination d in the pre- and postrepeal periods, respectively. Then, for destination d , we define $ACRI_d \equiv (\overline{RI}_{DAL,d,post} - \overline{RI}_{DAL,d,pre}) - (\overline{RI}_{HOU,d,post} - \overline{RI}_{HOU,d,pre})$ to measure the change in the average RI for DAL-origin passengers after adjusting for the average change in RI for HOU-origin passengers. Hereinafter, we refer to $ACRI_d$ as the *adjusted change in route inefficiency* for destination d . The mean (median) $ACRI_d$ is -0.2613 (-0.2506), indicating that DAL-origin passengers had access to more efficient routes to many destinations after the repeal.

We add the interactions of $ACRI_{d(i)}$ (i.e., the adjusted change in route inefficiency for passenger i 's destination) with $Dallas_i$, $Post_i$, and $Treated_i$ to Equation (1) to specify Model 3. Table 4 shows that the coefficient of $ACRI$ in Model 3, α_{12} , is positive and statistically significant, supporting our prediction that the destinations with larger adjusted decreases in route inefficiency benefited more from the repeal. For example, Model 3 suggests that a destination with an $ACRI$ value of -0.2506 (the median $ACRI$ across all destinations) would have a 50-minute decrease (i.e., $\hat{\alpha}_3 + \hat{\alpha}_{12} \times ACRI_d = -17.67 + 128.73 \times (-0.2506) = -49.93$) in STT . Section 3.3 performs destination-level analysis to generate deeper insights into the relationship between $ACRI_d$ and STT .

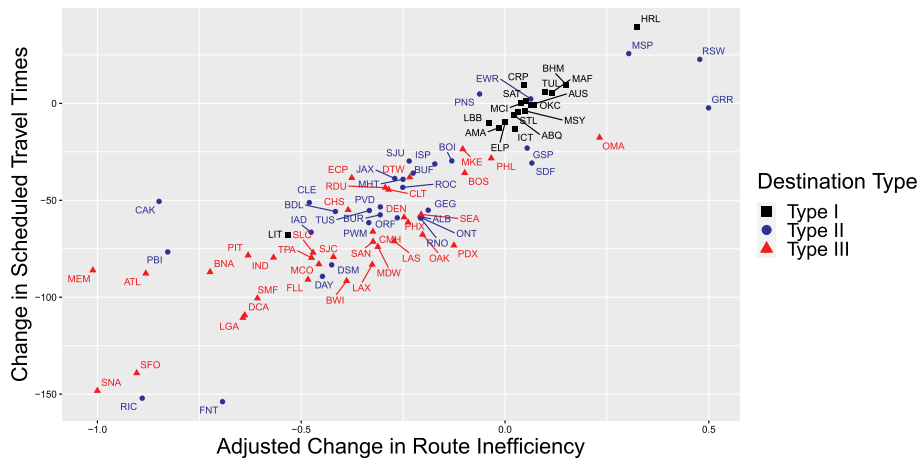
3.3. Destination-Level Analysis

In this section, we conduct our analysis in two steps. In the first step, we run a difference-in-differences model for each destination. To this end, let $\mathcal{I}_d$ denote the set of passengers who traveled from DAL and HOU to destination d . Then, for $i \in \mathcal{I}_d$, we run the following model:

$$STT_i = \alpha_{d0} + \alpha_{d1} Dallas_i + \alpha_{d2} Post_i + \alpha_{d3} Treated_i + \gamma_d \mathbf{Z}_i + v_i, \quad (3)$$

where $\mathbf{Z}_i$ is a column vector that includes control variables, γ_d is a row vector that represents the coefficients of those control variables, and v_i is the error term. Unlike our main model (Equation (1)), this model excludes the destination fixed effect and the nonstop distance because those variables would not be identified in a destination-level model. In the second step, we compare the adjusted change in a destination's route inefficiency score (i.e., $ACRI_d$) with the change in its STT due to the repeal (i.e., $\hat{\alpha}_{d3}$). Figure 3 illustrates this comparison and reveals that the efficiency changes show wide variations across destinations. Our route inefficiency metric

Figure 3. (Color online) Change in Scheduled Travel Times by Destination



explains this heterogeneity by linking the changes in scheduled travel times to the changes in route inefficiency scores.

We illustrate the usefulness of our route inefficiency metric by revisiting the motivating examples we provide in Table 2. In particular, our destination-level analysis reveals that $\hat{\alpha}_{MSP,3} = 25.89$ ($p < 10^{-10}$), indicating that the repeal led to a 25.89-minute increase in the average *STT* for passengers traveling from DAL to MSP. Our route inefficiency metric explains the mechanism behind this increase. Specifically, Southwest did not have a nonstop flight from HOU to MSP during our study period. As such, passengers traveling from HOU to MSP used Atlanta (ATL), DEN, MDW, MCI, and STL as the main layover points, both before and after the repeal. By contrast, passengers traveling from DAL to MSP had two main layover options before the repeal: STL and MCI. The nonstop distance between DAL and MSP is 853 miles, whereas the total distance the passenger travels is 994 miles with a layover in STL and 853 miles with a layover in MCI. Furthermore, STL and MCI are relatively less congested airports. In the postrepeal period, Southwest did not introduce nonstop service from DAL to MSP but began using ATL, DEN, and MDW as additional layover points. These airports not only increase travel distance (e.g., the total trip length from DAL to MSP with a layover in ATL is 1,628 miles) but also are more congested. Indeed, $\bar{R}_{DAL,MSP,pre} = 1.75$ and $\bar{R}_{HOU,MSP,pre} = 2.22$, which imply that during the prerepeal period, the routes connecting DAL and MSP were more efficient than those connecting HOU and MSP. The route-level changes after the repeal result in $ACRI_{MSP} = 0.31$, which explains the increase in *STT*.

For passengers traveling to ALB, we have $ACRI_{ALB} = -0.21$ and $\hat{\alpha}_{ALB,3} = -58.35$ with $p < 10^{-10}$. Intuitively, to travel from DAL to ALB during the prerepeal period, Southwest passengers relied on two layovers—one in an exempt destination, such as MCI, and another in a

large airport, such as Baltimore/Washington International Airport (BWI) or MDW. Although Southwest did not introduce a nonstop flight from DAL to ALB, eliminating one of the layovers improved Southwest's route efficiency, leading to a decrease in *STT*. Among type III destinations, both SNA and MDW benefited from the introduction of nonstop flights. However, the reduction in *STT* was much larger for passengers traveling to SNA. This is because many SNA passengers relied on two layovers during the prerepeal period, whereas MDW passengers relied on a single layover. Our route inefficiency metric captures this difference as we have $ACRI_{SNA} = -1.01$ and $ACRI_{MDW} = -0.31$, which translate to $\hat{\alpha}_{SNA,3} = -146.98$ ($p < 10^{-10}$) and $\hat{\alpha}_{MDW,3} = -73.77$ ($p < 10^{-10}$).

In summary, our route inefficiency metric, which combines travel distances with airport congestion, captures the heterogeneous impact of the route-level changes on different destinations. As expected, introducing nonstop service between o and d decreases the scheduled travel times between these locations. That said, our metric explains why some type III destinations had higher efficiency improvements than others. Our metric also captures the impact of the repeal on type I destinations, which were exempt from the repeal, and type II destinations to which Southwest did not start nonstop service after the repeal. For example, after launching nonstop service from o to d , the airline can use d as a layover airport to offer more efficient connecting flights from point o to a type II destination in its network (e.g., from DAL to ALB with a layover in MDW). Our metric captures this possibility by comparing travel distances and airport congestion on the routes the airline used before and after the repeal. Indeed, our metric's ability to parsimoniously capture the mechanism behind the changes in scheduled travel times suggests that airline managers and regulators can use it to assess the efficiency impact of future route-level changes.

4. Resilience Analysis

4.1. Analysis by Destination Type

In our resilience analysis, we follow the approach introduced in Section 3 and first run the following model:

$$\text{Delayed}_i = \alpha_{0,d(i)} + \alpha_1 \text{Dallas}_i + \alpha_2 \text{Post}_i + \alpha_3 \text{Treated}_i + \gamma \mathbf{X}_i + \epsilon_i, \quad (4)$$

where Delayed_i is a binary variable that equals one if passenger i 's delay exceeds three hours and zero otherwise. Although Equation (4) has a binary dependent variable, we treat it as a linear model because the marginal effects in a linear model are easier to interpret compared with nonlinear models, such as a logit model (e.g., Phillips and Sertsios 2013, Chen et al. 2021). Nevertheless, we reestimate Equation (4) as a logit model in Section B.5 in the online appendix to show the robustness of our findings. Section B.5 also presents models with an alternative delay cutoff (i.e., delays exceeding two hours) and quantile regression models as additional robustness checks.

Table 5 shows our coefficient estimates. The treatment effect coefficient in Model 4, α_3 , indicates that the repeal led to a 0.93-percentage-point decrease in a DAL-origin passenger's likelihood to experience a long delay, on average. Given that 2.5% of DAL-origin passengers experienced a long delay during the prerepeal period (as documented in Section 2.3), this estimate suggests a major overall resilience improvement for DAL-origin passengers. Next, we add the interactions of TypeII_i and TypeIII_i with Dallas_i , Post_i , and Treated_i to Equation (4) to specify Model 5, which estimates the resilience impact of the repeal on the three destination types. In Model 5, the α_3 coefficient indicates that the repeal led to a 0.19-percentage-point increase in a passenger's likelihood to experience a long delay while traveling from DAL to a type I destination. In addition, $\hat{\alpha}_3 + \hat{\alpha}_8 = 0.0019 - 0.0145 = -0.0126$ and $\hat{\alpha}_3 + \hat{\alpha}_9 = 0.0019 - 0.0222 = -0.0203$ indicate that the repeal led to 1.26- and 2.03-percentage-point decreases in a passenger's likelihood to experience a long delay while traveling from DAL to type II and type III destinations, respectively. Last, comparing α_8 and α_9 in Model 5 with an F -test reveals a statistically significant difference ($p < 10^{-10}$) between the resilience impact of the repeal on type II and type III destinations.

Our findings suggest that eliminating at least one leg and one layover improved resilience for passengers traveling from DAL to type III destinations. The mechanism behind the repeal's negative impact on type I destinations and positive impact on type II destinations is less clear. In particular, although Southwest offered nonstop flights to type I destinations both before and after the repeal, the repeal increased the likelihood of long delays for passengers traveling from DAL to those destinations. Furthermore, although type II destinations relied on connections both before and after the repeal,

Table 5. Resilience Regressions

	Model 4	Model 5	Model 6
Intercept	0.03605*** (0.00057)	0.03815*** (0.00058)	0.03987*** (0.00058)
Nonstop distance	−0.00002*** (0.000001)	−0.00001*** (0.000001)	−0.00002*** (0.000001)
Min. temperature	−0.00061*** (0.00001)	−0.00059*** (0.00001)	−0.00059*** (0.00001)
Max. wind	0.00092*** (0.00001)	0.00089*** (0.00001)	0.00090*** (0.00001)
Max. precipitation	0.02034*** (0.00027)	0.02031*** (0.00027)	0.02039*** (0.00027)
Min. visibility	−0.00114*** (0.00001)	−0.00111*** (0.00001)	−0.00112*** (0.00001)
Max. congestion	0.01582*** (0.00030)	0.01160*** (0.00030)	0.01248*** (0.00030)
Dallas, α_1	0.00717*** (0.00014)	−0.00345*** (0.00016)	0.00443*** (0.00014)
Post, α_2	−0.00295*** (0.00010)	−0.00174*** (0.00016)	−0.00261*** (0.00011)
Treated (Dallas × Post), α_3	−0.00929*** (0.00015)	0.00188*** (0.00020)	−0.00657*** (0.00016)
Dallas × Type II, α_4		0.01429*** (0.00059)	
Dallas × Type III, α_5		0.02096*** (0.00026)	
Post × Type II, α_6		−0.00443*** (0.00039)	
Post × Type III, α_7		−0.00115*** (0.00021)	
Treated × Type II, α_8		−0.01453*** (0.00071)	
Treated × Type III, α_9		−0.02221*** (0.00032)	
Dallas × ACRR, α_{10}			0.10147*** (0.00154)
Post × ACRR, α_{11}			−0.00687*** (0.00113)
Treated × ACRR, α_{12}			−0.10963*** (0.00181)
Observations	12,464,798	12,464,798	12,412,962
Adjusted R^2	0.01534	0.01650	0.01627

Notes. Standard errors clustered at the destination level are in parentheses. The dependent variable takes the value one if a passenger experiences a delay of three hours or longer and zero otherwise. All three models have destination, month, and day of the week fixed effects. We do not report the fixed effect coefficients because of space limitations.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

the likelihood of long delays for passengers traveling from DAL to those destinations decreased. Therefore, we cannot attribute the impact of the repeal solely to Southwest's ability to offer nonstop flights to all destinations in its network. In the next section, we introduce a route resilience metric to shed light on the heterogeneous impact of the repeal on different destinations.

4.2. The Role of Route Resilience

Both academics and practitioners emphasize the importance of buffers for resilience in the airline industry (e.g., Wu and Caves 2000, Stalnaker et al. 2019). In particular,

studies using flight-level data show the benefit of schedule buffers in reducing flight delays (e.g., Ahmadbeygi et al. 2010, Deshpande and Arıkan 2012, Arıkan et al. 2013, Brueckner et al. 2021). Furthermore, more frequent flights between an origin-destination pair serve as a capacity buffer, which affects how fast an airline can reallocate a passenger experiencing a flight disruption (e.g., Barnhart et al. 2014, Zou and Hansen 2014). We operationalize these two factors as follows: For leg l in passenger i 's itinerary, recall that $flights_{il}$ denotes the number of nonstop flights from the origin to the destination of that leg on passenger i 's travel date. Then, we define the route resilience for passenger i as

$$RR_i = \frac{STT_i}{NomTT_i} \prod_{l=1}^{L_i} \frac{flights_{il}}{1 + flights_{il}}, \quad (5)$$

where STT_i and $NomTT_i$ denote passenger i 's scheduled and nominal travel times, respectively.

In Equation (5), a longer buffer in a passenger's itinerary leads to a larger $STT_i/NomTT_i$ ratio, which, in turn, increases the passenger's RR score. Similarly, higher flight frequencies along a passenger's route lead to higher $flights_{il}/(1 + flights_{il})$ ratios, which also lead to a higher RR score. All else being equal, a passenger's RR score decreases as the number of legs in the itinerary increases. This is because $flights_{il}/(1 + flights_{il}) \in (0, 1)$, which implies that adding a new leg will multiplicatively adjust the route resilience formula by a positive scalar that is less than one.⁵

We predict that the destinations with large increases in the RR values for DAL-origin passengers would experience larger resilience improvements. To test this prediction, we let $\overline{RR}_{o,d,pre}$ and $\overline{RR}_{o,d,post}$ denote the average RR values for passengers who traveled from origin $o \in \{DAL, HOU\}$ to destination d during the pre- and postrepeal periods, respectively. Accordingly, for destination d , we define $ACRR_d \equiv (\overline{RR}_{DAL,d,post} - \overline{RR}_{DAL,d,pre}) - (\overline{RR}_{HOU,d,post} - \overline{RR}_{HOU,d,pre})$. This metric, which we refer to as the *adjusted change in route resilience*, measures the change in the average RR for DAL-origin passengers after adjusting for the average change in RR for HOU-origin passengers. The mean and median $ACRR_d$ are 0.0577 and 0.0369, respectively, implying that DAL-origin passengers had access to more resilient routes to many destinations after the repeal. We add the interactions of $ACRR_{d(i)}$ (i.e., the adjusted change in route resilience for passenger i 's destination, $d(i)$) with $Dallas_i$, $Post_i$, and $Treated_i$ to Equation (4) to specify Model 6.

As Table 5 shows, the $ACRR$ coefficient in Model 6, α_{12} , is negative and statistically significant, supporting our prediction that the destinations with larger increases in route resilience benefited more from the repeal. For example, Model 6 indicates that a passenger's likelihood of experiencing a long delay while

traveling from DAL to a destination with an $ACRR$ value of 0.0369 (the median $ACRR$ across all destinations) would be 1.06 percentage points lower (i.e., $\hat{\alpha}_3 + \hat{\alpha}_{12} \times ACRR_d = -0.0066 - 0.1096 \times 0.0369 = -0.0106$) after the repeal. Section 4.3 performs destination-level analysis to more closely examine the relationship between $ACRR_d$ and long delays.

4.3. Destination-Level Analysis

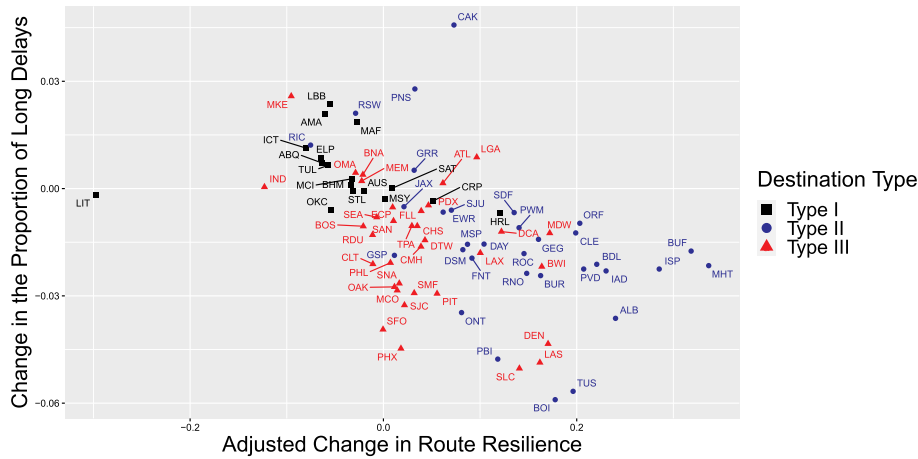
In this section, we follow the two-step procedure introduced in Section 3.3 to generate deeper insights into the role of route-level changes in long delays. In the first step, we run the following model for the set of passengers who traveled from DAL and HOU to destination d :

$$Delayed_i = \alpha_{d0} + \alpha_{d1} Dallas_i + \alpha_{d2} Post_i + \alpha_{d3} Treated_i + \gamma_d Z_i + v_i. \quad (6)$$

In the second step, we compare the adjusted change in each destination's route resilience score (i.e., $ACRR_d$) with the change in long delays due to the repeal (i.e., $\hat{\alpha}_{d3}$). Figure 4 visualizes this comparison and generates two takeaways. First, the repeal has a heterogeneous impact across destinations, where the destination type does not sufficiently explain the mechanism behind the observed heterogeneity. Second, our route resilience metric better explains the observed heterogeneity by linking the changes in long delays to the changes in route resilience scores.

Recall from Section 3.3 that the repeal led to a 26-minute increase in STT for passengers traveling from DAL to MSP. Changing the layover points from smaller airports, such as MCI and STL, to busier airports, such as ATL, DEN, and MDW, was the main driver of this increase. Our destination-level resilience analysis reveals that the repeal was not all bad news for passengers traveling from DAL to MSP. Specifically, we find that Southwest had higher flight frequencies in routes with busier layover airports (e.g., DAL-MDW-MSP), leading to $ACRR_{MSP} = 0.087$. This resilience improvement was associated with a $\hat{\alpha}_{MSP,3} = 1.54$ ($p = 0.0057$)-percentage-point decrease in the likelihood of a long delay while traveling from DAL to MSP.

Our destination-level resilience analysis also reveals why the likelihood of a long delay increased for passengers traveling from DAL to type I (exempt) destinations. In particular, Southwest reduced scheduled travel times (and, therefore, schedule buffers) of flights from DAL to those airports, as Model 2 in Section 3.1 shows. Southwest also reduced flight frequencies from DAL to most of those airports because the number of passengers using those airports for layovers sharply declined after the repeal. For example, the daily average number of flights from DAL to ABQ was 7.42 during the prerepeal period and 4.88 during the postrepeal period. Combining

Figure 4. (Color online) Change in the Proportion of Long Delays by Destination

a decrease in buffer times with a reduction in flight frequencies led to a decline in route resilience scores, which in turn is associated with an increase in the likelihood of a long delay.

In summary, our route resilience metric explains the mechanism behind the heterogeneous impact of the repeal across destinations. Specifically, starting nonstop service from o to d improves the resilience of an airline's operations for the o - d pair by eliminating a flight leg and a layover. Notably, the magnitude of the resilience improvement depends on the connection airport(s) the airline used to transport passengers from o to d before the introduction of the nonstop flight. This is because the connection airport—airport c —affects an itinerary's schedule buffer and the ease of reaccommodating a passenger in case of a missed connection. Moreover, launching nonstop service from o to d can adversely affect airport c as the airline can reduce flight frequencies and scheduled flight times from o to c because of its reduced reliance on c for connections. Our route resilience metric parsimoniously captures these route-level dynamics and explains why some destinations benefited from the route-level changes triggered by the repeal whereas others suffered. Furthermore, a key insight from our efficiency and resilience analyses is that focusing on one metric (i.e., efficiency or resilience) in isolation may not offer a complete view of the operational impact of route-level changes. Indeed, our results imply that some route-level decisions (e.g., reducing itinerary buffers) warrant further attention because they lead to tradeoffs between efficiency and resilience.

5. Managing the Efficiency-Resilience Tradeoff

The airline studies that use flight-level data to quantify the role of air and ground buffers in flight delays (e.g., Ahmadbeygi et al. 2010, Arkan et al. 2013, Kafle and

Zou 2016, Brueckner et al. 2021) are informative for passengers taking nonstop flights. However, they do not fully capture connecting passengers' travel experiences because of the lack of itinerary-level data. By contrast, our data set enables us to examine the relationship between efficiency and resilience for connecting passengers. In particular, our analysis in Section 4 illustrates that itinerary buffers (measured by $STT_i/NomTT_i$) play an important role in resilience. Thus, one way to improve resilience is to increase scheduled travel times. However, increasing scheduled travel times harms efficiency.

In this section, we analyze the efficiency-resilience tradeoff for connecting passengers by focusing on the two-leg itineraries originating from DAL during the postrepeal period.⁶ Let $\mathcal{I}_r$ denote the set of passengers traveling through route r during the postrepeal period. For example, $\mathcal{I}_{DAL-MCI-MDW}$ is the set of passengers who traveled from DAL to MDW through a layover in MCI during the postrepeal period. For a given route r and itinerary buffer threshold k , we first define two mutually exclusive and exhaustive subsets of $\mathcal{I}_r$ as follows: $\hat{\mathcal{I}}_r^k \equiv \{i \in \mathcal{I}_r : STT_i/NomTT_i < k\}$ and $\tilde{\mathcal{I}}_r^k \equiv \{i \in \mathcal{I}_r : STT_i/NomTT_i \geq k\}$. Here, $\hat{\mathcal{I}}_r^k$ is the set of high-risk itineraries in terms of their propensity to experience long travel delays. We reallocate a passenger with a high-risk itinerary to an alternative itinerary that covers the same route on the same day with an itinerary buffer that exceeds k . If multiple alternative itineraries satisfy our criteria, we pick the one with the shortest time difference from the passenger's original scheduled departure time.⁷ The flight frequency component of our route resilience metric plays a role here because it is easier to find an alternative itinerary when the flight frequencies between the origin-destination pairs in an itinerary are high. If no itineraries satisfy our criteria, we retain the original itinerary.

After reallocating all high-risk passengers to alternative itineraries, we recalculate the total *STT* and travel delays for passengers traveling through route r . The next step in our analysis is to pick the best k for each route. Note that picking a larger k will require the airline to reallocate more passengers to alternative itineraries with longer buffers. On the one hand, these reallocations can increase scheduled travel times. On the other hand, they can lead to reductions in travel delays. Hence, for each route r , we assume that the airline picks the best k that minimizes a weighted average of scheduled travel times and travel delays. Formally, for route r , we solve

$$\min_k \underbrace{\sum_{i \in \mathcal{I}_r^k} (STT_{j(i)}(k) + \gamma_r \text{Delay}_{j(i)}(k))}_{\text{reallocated passengers}} + \underbrace{\sum_{i \in \mathcal{I}_r^k} (STT_i(k) + \gamma_r \text{Delay}_i(k))}_{\text{passengers without reallocation}}, \quad (7)$$

where $j(i)$ denotes the itinerary to which we reallocate itinerary i and γ_r represents the relative importance of delays for the airline in route r .⁸ For a given γ_r , we numerically find the optimal k for route r , $k_r^*(\gamma_r)$, with a simple line search, where we vary k between 1 and 1.25 in increments of 0.01. In our counterfactual analysis, we treat γ_r as a model parameter and obtain results for a range of γ_r values. In practice, airline managers can determine γ_r based on airline and route characteristics (e.g., point-to-point versus hub-and-spoke, aircraft type). In general, having $\gamma_r \geq 1$ would be reasonable because delays create additional costs for airlines, for example, extra crew costs and passenger compensation for missed connections (e.g., Ferguson et al. 2013).

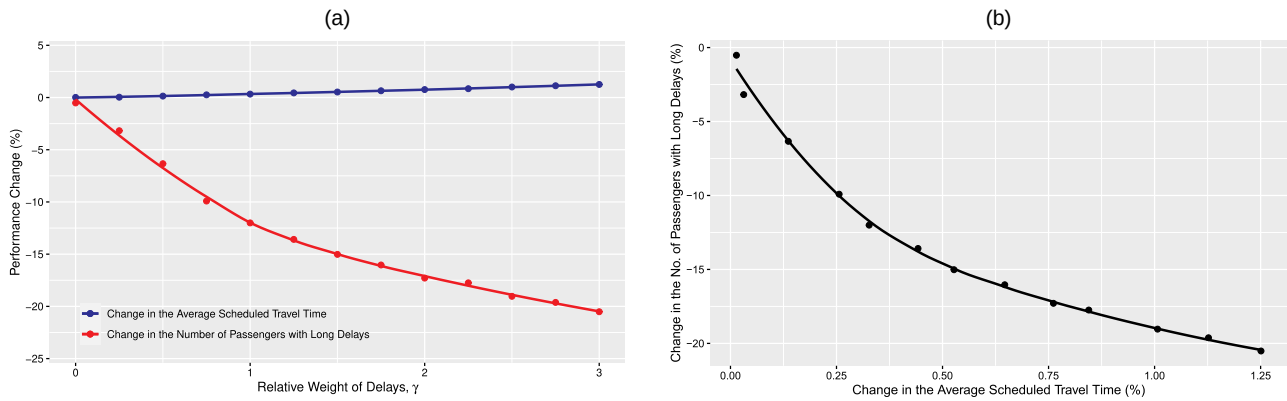
Let $\mathcal{R}$ denote the set of two-leg DAL-origin routes during the postrepeal period. For a given γ_r , we calculate $\sum_{r \in \mathcal{R}} \sum_{i \in \mathcal{I}_r} STT_i$ and $\sum_{r \in \mathcal{R}} (\sum_{i \in \mathcal{I}_r^{k_r^*(\gamma_r)}} STT_{j(i)}(k_r^*(\gamma_r)) + \sum_{i \in \mathcal{I}_r^k} STT_i(k_r^*(\gamma_r)))$, where the former denotes the total *STT* for all two-leg passengers traveling from DAL to other destinations during the postrepeal period and the latter denotes the same metric after we reallocate the high-risk passengers to alternative itineraries based on the route-specific optimal itinerary buffer thresholds, $k_r^*(\gamma_r)$.⁹ We use these two metrics to calculate the percentage change in the total *STT* and the number of passengers experiencing long delays as a function of γ_r .

In our data set, about 775,000 DAL-origin passengers had two-leg itineraries during the postrepeal period. This number corresponds to 20.4% of all DAL-origin passengers during the postrepeal period. The average *STT* and travel delay for those passengers were 278.7 minutes and 24.1 minutes, respectively. In addition, 13,869 passengers experienced long delays (i.e., delays exceeding three hours). Figure 5(a) illustrates the changes in the average *STT* and the number of

passengers with long delays as a function of γ_r , where we set $\gamma_r = \gamma$ for all $r \in \mathcal{R}$ and vary γ between 0 and 3 in increments of 0.25. This figure reveals that the average *STT* increases and the number of passengers experiencing long delays decreases in γ . For example, when $\gamma_r = 1$ for all $r \in \mathcal{R}$, the airline would reallocate 14,191 passengers (which is about 2% of approximately 775,000 passengers we consider) to alternative itineraries. Consequently, the average *STT* would increase by 0.33% (0.9 minutes), and the average delay would decrease by 8.5% (2.1 minutes). Furthermore, the number of passengers experiencing long delays would decrease by 12% (from 13,869 to 12,204). By contrast, when $\gamma_r = 2$ for all $r \in \mathcal{R}$, the airline would reallocate 30,080 passengers to alternative itineraries. Consequently, the average *STT* would increase by 0.76%, and the number of passengers experiencing long delays would decrease by 17%. Intuitively, a larger γ value requires the airline to reallocate more passengers, which leads to a larger increase in the average scheduled travel time and a larger decrease in the number of passengers with long delays.

Figure 5(b) further illustrates the efficiency-resilience tradeoff by showing the change in the number of passengers with long delays as a function of the change in the average *STT*. This figure reveals an efficient frontier for the relationship between efficiency and resilience. Specifically, airline managers can specify the maximum efficiency decrease they can tolerate, determining the resilience improvement they can obtain in return. For example, if a manager is willing to increase the average *STT* by 0.5% by reallocating high-risk passengers to alternative itineraries, they can decrease the number of passengers experiencing long delays by about 14%.

The main message emerging from our counterfactual analysis is that small but carefully planned increases in scheduled travel times can significantly reduce the number of passengers with long delays. We convey this message by reallocating high-risk itineraries to alternative itineraries with longer buffers. In practice, it would be impractical for an airline to sell high-risk itineraries first and then change them. Hence, the counterpart of our procedure for an airline would be to prevent passengers from purchasing high-risk itineraries in the first place. Although many studies focus on optimizing flight schedules to reduce delays, our approach suggests that airlines can reduce travel delays by carefully curating the itineraries they offer to their connecting passengers. Indeed, our suggestion is easier to implement as it does not require changing flight schedules. Admittedly, the actual magnitudes of the efficiency and resilience changes may be larger or smaller in practice as passengers may pick alternative itineraries with shorter or longer scheduled travel times and itinerary buffers than the ones our procedure picks for them because of prices, seat unavailability, etc. Nevertheless, our findings highlight the importance of itinerary buffers and carefully

Figure 5. (Color online) The Efficiency-Resilience Tradeoff for Connecting Passengers

Notes. We set $\gamma_r = \gamma$ for all $r \in \mathcal{R}$ and vary γ between 0 and 3 in increments of 0.25. Panel (a) shows the changes in the average STT and the number of passengers experiencing long delays as a function of γ . Panel (b) uses the same data but places the change in the average STT on the horizontal axis and the change in the number of passengers experiencing delays on the vertical axis.

selecting the set of itineraries available for purchase to manage the efficiency-resilience tradeoff for connecting passengers.

6. Robustness Checks

6.1. Parallel Trends Assumption

Our models rely on the assumption that the time trends for treated and untreated passengers (i.e., DAL- and HOU-origin passengers, respectively) would be parallel in the absence of the repeal. One approach to test the validity of the parallel trends assumption is to run separate models for DAL- and HOU-origin passengers with time fixed effects to check whether those fixed effects show similar patterns. As such, we run a separate model for each origin-destination pair, where the explanatory variables include year-week fixed effects and the control variables of our main models. Our two-year window leads to 107 year-week values, starting with the 41st week of 2013 and ending with the 41st week of 2015. In our models, the intercept captures the 41st week of 2013. In addition, the 106 year-week fixed effect coefficients capture the incremental impact of those weeks, where the first 52 year-week fixed effect coefficients cover the prerepeal period.

Figure 6(a) shows the year-week fixed effect coefficients along with their 95% confidence intervals (CIs) for the scheduled travel times of the HOU- and DAL-origin passengers traveling to MDW. For each year-week before the repeal, we find that the 95% CIs of the year-week fixed effects of the models we run for HOU- and DAL-origin passengers overlap. This finding indicates that weekly fluctuations in STT follow similar patterns for HOU- and DAL-origin passengers traveling to MDW. We repeat the same procedure for all destinations and count the number of overlapping CIs for each destination. The mean and median numbers of overlapping CIs across all destinations are 50.04 and 52,

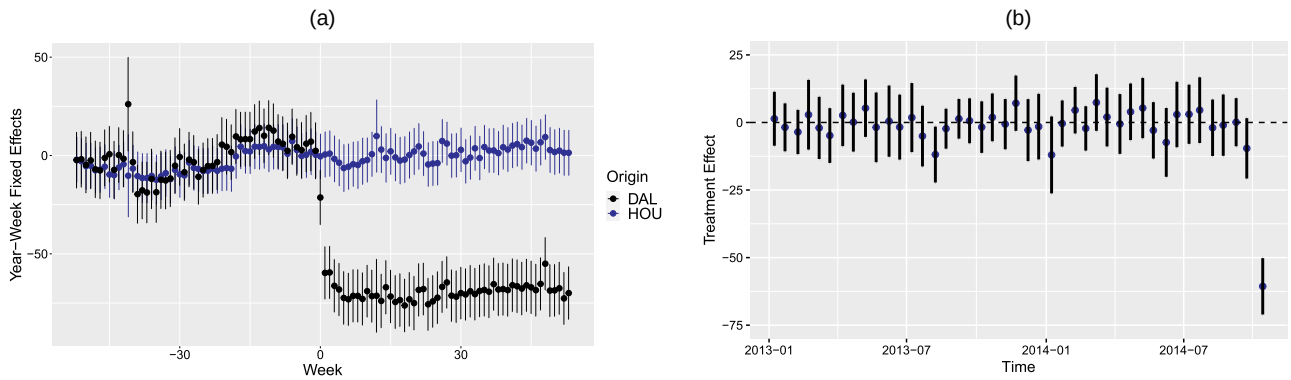
respectively. In other words, many destinations have fully overlapping 95% CIs for the year-week fixed effects before the repeal. Applying the same procedure to our resilience analysis reveals that the mean and median numbers of overlapping CIs across all destinations are 50.73 and 52, respectively. These numbers provide empirical support for the parallel trends assumption in our efficiency and resilience analysis.

6.2. Placebo Tests

In this section, we conduct placebo tests to further check whether any systematic differences between DAL and HOU confound our results. Our large data set enables us to perform 42 placebo tests on an operational performance metric (efficiency or resilience) for each destination. Specifically, we create a pseudo-event on the 8th and 22nd day of each month between January 2013 and September 2014. For each pseudo-event, we use the seven-day window before the event as the preevent period and the seven-day window after the event as the postevent period. For example, in our first placebo test, the time between January 1, 2013, and January 7, 2013, is the pre-event period, and the time between January 8, 2013, and January 14, 2013, is the postevent period. Similarly, in our second placebo test, the time between January 15, 2013, and January 21, 2013, is the pre-event period, and the time between January 22, 2013, and January 28, 2013, is the postevent period.

In our efficiency analysis, we estimate Equation (3) for each placebo event. For comparison purposes, we also estimate the impact of the actual repeal by using the time between October 6, 2014, and October 12, 2014, as the pre-event period and the time between October 13, 2014, and October 19, 2014, as the postevent period. Figure 6(b) shows the treatment effect coefficients and their 95% CIs for the 42 placebo events and the actual event for MDW. Only one pseudo-event leads to a

Figure 6. (Color online) Parallel Trends and Placebo Events



Notes. In panel (a), each vertical bar represents a year-week fixed effect for DAL- or HOU-origin passengers traveling to MDW. In panel (b), each vertical bar except the last one represents a pseudo-event for DAL- and HOU-origin passengers traveling to MDW, whereas the last vertical bar represents the actual repeal.

statistically significant treatment effect. By contrast, $\hat{\alpha}_{MDW,3} = -62$ minutes ($p < 10^{-16}$) for the actual repeal, which implies that the repeal led to an immediate reduction in *STT*.

Repeating the same procedure for all destinations in our data set leads to $42 \times 87 = 3,654$ pseudo-events. The average treatment effect of those pseudo-events is 0.81 minutes, which is nonsignificant at $p = 0.05$. By contrast, the average treatment effect of the actual events is -22 minutes, which is statistically significant ($p < 10^{-5}$). We repeat the same procedure on long delays. Specifically, for each destination, we estimate Equation (6) for 42 placebo events and the actual event. The average treatment effect of those pseudo-events is -0.02 percentage points (nonsignificant at $p = 0.05$). By contrast, the average treatment effect of the actual event in a two-week window is -3.07 percentage points ($p < 10^{-5}$). Thus, we conclude that HOU-origin passengers are a suitable control group in studying the impact of the repeal on DAL-origin passengers.

6.3. Potential Violations of SUTVA

Our analysis relies on SUTVA, which requires that the repeal not affect the control group (i.e., HOU-origin passengers). Although the HOU-origin passengers were not subject to the Wright Amendment, its repeal could indirectly impact them by changing Southwest's propensity to use HOU (DAL) as a layover airport for its DAL-origin (HOU-origin) passengers. To this end, Section B.1 in the online appendix identifies the destinations most prone to a potential violation of SUTVA because of Southwest's routing decisions. For example, ATL is one such destination, as 65% of passengers traveling from DAL to ATL used HOU as a layover airport during the prerepeal period, whereas only 8% of DAL-origin passengers traveling to ATL had a layover in HOU during the postrepeal period. In Section B.1, we

perform subsample analyses and show that our findings continue to hold on subsamples excluding all passengers traveling from HOU and DAL to destinations most susceptible to a potential SUTVA violation. See Section B.1 in the online appendix for details.

6.4. Other Robustness Checks

In Section B in the online appendix, we present additional robustness checks. Specifically, Section B.2 fits separate models for each destination type to check the robustness of Models 2 and 5. Section B.3 analyzes two alternative time windows, one-year and 18-month periods surrounding the repeal. Section B.4 uses actual travel times (i.e., scheduled travel times plus travel delays) as a proxy for efficiency. Section B.5 checks the robustness of our resilience analysis by (i) using a binary dependent variable that equals one if a passenger's delay exceeds two hours and zero otherwise, (ii) reestimating Models 4–6 as logit models, and (iii) treating delays as a continuous variable and analyzing quantile regression models. Section B.6 controls for airline competition. Last, Section B.7 provides additional counterfactual analysis. Collectively, these analyses show the robustness of our findings. See Section B in the online appendix for details.

7. Discussion and Conclusion

We study the role of an airline's route-level decisions in the efficiency and resilience of its operations. Unlike the relevant literature, which typically relies on publicly available flight-level (OTP) data, our study uses a passenger-level data set, allowing us to better capture passengers' travel experiences by incorporating scheduled and actual flight and layover times into the analysis. Our contributions are threefold. First, we leverage the Wright Amendment repeal to quantify the impact of major route-level changes on passenger travel times and

delays. Our analysis reveals that the impact of the repeal on both efficiency and resilience varies widely across destinations. Initially, it might be assumed that Southwest's decision to start nonstop service to some destinations but not to others would explain these variations. However, our findings show that this decision alone does not sufficiently explain the observed heterogeneity.

Second, we develop two route-level metrics—route inefficiency and route resilience. Our metrics parsimoniously capture the role of route characteristics (i.e., travel distances, airport congestion, flight frequencies, and itinerary buffers) and thereby explain the heterogeneous impact of the repeal across destinations. Finally, our counterfactual analysis shows that limiting passengers' access to high-risk itineraries with short buffer times can lead to major resilience improvements without significantly deteriorating efficiency. Our analysis translates to identifying high-risk itineraries and preventing passengers from purchasing them. Notably, a novel feature of this approach is that it does not require any changes in flight schedules. Hence, our approach may be easier to implement than alternative approaches requiring airlines to reconfigure their flight schedules.

Although we rely on the route-level changes triggered by the Wright Amendment repeal to generate our findings, our study has broader implications for managers, regulators, and researchers as airline networks constantly evolve, albeit more slowly than the period we study. For managers, our analysis suggests that focusing on a single metric (i.e., efficiency or resilience) in isolation may not provide a complete view of the operational impact of route-level changes. This is because some changes (e.g., the introduction of nonstop service) create synergies between efficiency and resilience, whereas others (e.g., the reduction in itinerary buffers) lead to tradeoffs. Thus, airline managers should consider both the efficiency and resilience impact of their decisions to obtain a more comprehensive view of the operational implications of such decisions. Additionally, our findings indicate that route-level changes (e.g., introducing nonstop service between two airports) can benefit some passengers (destinations) while disadvantaging others by triggering other changes (e.g., reducing flight frequencies to airports previously used for connections). Therefore, airline managers should look beyond a focal origin-destination pair to have a more complete assessment of the impact of a route-level decision on their entire passenger pool.

For managers and regulators, our study highlights the monetary value of passenger travel time reductions. Such a value is especially important for the FAA in our context because of its goal to assess travel time savings or losses due to regulatory actions (FAA 2020). Our analysis reveals that passengers who flew from DAL to type III destinations saved 79 minutes, on average. Using the \$36.10 figure the FAA provides as the hourly

value of travel time savings for personal travel, we find that the time savings of approximately two million passengers who flew from DAL to type III destinations in the one-year period after the repeal translates to $2 \text{ million passengers} \times 79/60 \text{ hours/passenger} \times \$36.10/\text{hour} = \$95 \text{ million annually}$. Considering approximately the same number of passengers who flew from the nonexempt destinations to DAL as additional beneficiaries, we conclude that in just one year, the repeal created a welfare gain of more than \$190 million because of reduced travel times. This conservative estimate, which does not consider several factors, such as the higher monetary value of travel time savings for business travelers, demonstrates how seemingly small travel restrictions can have significant welfare implications. Although the Wright Amendment repeal eliminated one such restriction, Reagan National Airport (DCA) and LaGuardia Airport (LGA) currently have similar perimeter rules restricting nonstop service from those airports to long-haul destinations. Our approach of predicting travel time savings using a route inefficiency metric can help regulators evaluate the impact of government regulations on passenger welfare.

Although Southwest does not have the best on-time performance record, it typically has shorter delays compared with other major airlines, such as United Airlines (e.g., Silver 2015). Furthermore, other major airlines rely more on connecting flights, which, as our analysis shows, are more prone to longer travel delays for passengers. Considering these factors, it is unclear which major airline offers its passengers the shortest travel times and best reliability. Thus, another practical implication of our study for regulators is that passenger-level data (e.g., passenger travel times, which include layover times for connecting passengers) can be more informative than flight-level data (e.g., on-time performance) to assess operational performance in the airline industry. In line with efforts to improve regulatory effectiveness (e.g., Johnson et al. 2023), the U.S. Department of Transportation (DOT) might consider releasing passenger travel time statistics as an additional performance metric.

DOT already publishes the Airline Origin and Destination Survey, which is a 10% sample of airline tickets from reporting carriers. This data set includes origin, destination, and layover airports but excludes travel times. In light of our findings, adding at least the scheduled travel times to this database would be helpful in tracking travel time patterns. Alternatively, DOT might consider publishing monthly on-time performance statistics at the passenger level (in addition to the flight level) so that the public can access statistics on the percentage of passengers who reach their final destinations on time. In line with the broader service operations literature (e.g., Buell et al. 2017, 2021), such transparency can incentivize airlines to pay more attention to passengers'

travel experiences and thereby improve service quality in the airline industry.

For researchers, the main implication of our study is the usefulness of our route efficiency and resilience metrics. In the absence of travel time data, future studies can use our efficiency and resilience metrics or their components (e.g., travel distances, flight frequencies) to incorporate route characteristics into operational performance analysis. Our efficiency and resilience metrics can also be used to assess (predict) the operational implications of current (future) route-level changes. For example, Southwest recently started service to Chicago's O'Hare International Airport (ORD). Historically, Southwest served only MDW, the secondary airport in Chicago. As such, it is unclear how entering ORD will affect the efficiency and resilience of Southwest's operations in Chicago. In particular, ORD is more congested than MDW. Thus, Southwest may need to allocate longer air and ground buffers in ORD, which might decrease Southwest's efficiency. Furthermore, serving some cities from ORD might require Southwest to reduce its flight frequencies to those cities from MDW. Hence, it will be interesting to observe if entering ORD will have a negative resilience spillover impact on Southwest's operations in MDW. We hope our study generates more interest in investigating and improving airline passengers' travel experiences.

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Endnotes

¹ As per the request of Southwest Airlines, we have the following disclosure regarding our data set: The "passenger-level" data referenced in the work do not include any personally identifiable passenger information, fares, etc.—only scheduled and actual flight times on a per-passenger basis.

² For round trips, our data set only includes the part of the trip originating from HOU or DAL. For example, if a passenger has a round trip between MDW and DAL, we only observe the return trip from DAL to MDW.

³ In our analysis, we exclude passengers flying between DAL and HOU because a control group does not exist for those passengers.

⁴ For example, suppose a passenger travels from DAL to Los Angeles (LAX) with a layover in ABQ. The distances between DAL and ABQ, ABQ and LAX, and DAL and LAX are 580, 677, and 1,246 miles, respectively. Also, suppose that the congestion at DAL at the passenger's departure is 0.2, the congestion at ABQ at arrival is 0.5, the congestion at ABQ at departure is 0.4, and the congestion at LAX at arrival is 0.8. Then, this passenger's route inefficiency score is $(580 + 677)/1,246 \times (1 + (0.2 + 0.5)/2) \times (1 + (0.4 + 0.8)/2) = 2.18$. Now consider a passenger who takes a nonstop flight from DAL to LAX. Suppose that the congestion levels at DAL at departure and LAX at arrival are identical to our previous example (0.2 and 0.8,

respectively). Then, this passenger's route inefficiency score is $1,246/1,246 \times (1 + (0.2 + 0.8)/2) = 1.5$, which is lower than the one we obtained in our first example.

⁵ We use two passengers from our data set to illustrate how *RR* works. The first passenger traveled from HOU to MSP with a layover in MDW. This passenger's *STT* and *NomTT* were 305 minutes and 243 minutes, respectively. On this passenger's travel date, Southwest had eight nonstop flights from HOU to MDW and six nonstop flights from MDW to MSP. The second passenger traveled from DAL to MSP with a layover at MCI. This passenger's *STT* and *NomTT* were 190 minutes and 171 minutes, respectively. On this passenger's travel date, Southwest offered seven nonstop flights from DAL to MCI and two nonstop flights from MCI to MSP. Note that the second passenger has a smaller buffer, 19 minutes ($190 - 171$), compared to the first passenger's 62 minutes ($305 - 243$), which increases the likelihood of a disruption (e.g., a missed connection). Furthermore, the second passenger's route has lower flight frequencies. Thus, the second passenger would be more likely to experience a long delay in case of a missed flight. Equation (5) captures these dynamics with a single metric. That is, the *RR* scores for the first and second passengers are $305/243 \times (8/(1+8)) \times (6/(1+6)) = 0.95$ and $190/171 \times (7/(1+7)) \times (2/(1+2)) = 0.65$, respectively.

⁶ Focusing on DAL-origin passengers in our counterfactual analysis eases exposition. Section B.7 in the online appendix shows that including passengers originating from HOU in the analysis does not change our insights.

⁷ When multiple alternative itineraries satisfy our criteria, other reallocation procedures, such as randomly picking an alternative itinerary or picking the alternative itinerary with the earliest departure time, lead to the same insights.

⁸ An alternative way to specify the objective function in Equation (7) is to make it a weighted combination of scheduled travel times and the number of passengers with long delays, where the weight assigned to *STT* would represent a penalty for each additional minute of *STT* and the weight assigned to the number of passengers with long delays would represent a penalty for each passenger experiencing a long delay. We use the objective function specified in Equation (7) because of its ease of interpretation.

⁹ For example, when $\gamma_r = 2$, the optimal k value is 1.13 for the DAL-MCI-MSP route and 1.2 for the DAL-ATL-MSP route (i.e., $k_{DAL-MCI-MSP}^*(2) = 1.13$ and $k_{DAL-ATL-MSP}^*(2) = 1.2$). That is, Southwest would have a higher reallocation threshold for passengers connecting through ATL.

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